Poker Exercises 3

- A poker hand contains 5 out of 52 cards. The cards are 13 kinds (1-Ace, 2,..,10,J,Q,K) and 4 suites (spades, hearts, clubs, diamonds)
- How many hands are flush, i.e., a flush consists of 5 card of the same suit, excluding the cases of royal and straight flush.
- The flush is a combination of 5 out of 13 cards of the same suit. Thus, there are $4 \cdot C(13,5)$ flushes. After excluding the straight and the royal flashes the number is $4 \cdot C(13,5)$-40
- How many hands contain 3-of a kind (e.g., 3 aces - the 4^{th} and 5^{th} card must be different because otherwise we would have stronger hands, to be discussed later):
- Number of ways to select the kind of the 3-of a kind C(13,1).
- Number of ways to select the suites of the 3-of a kind C(4,3).
- Number of ways to select the kind of the 4^{th} and 5^{th} card: C(12,2).
- Number of ways to select the suit of the 4^{th} and 5^{th} card: $C(4,1) \cdot C(4,1)$
- Total number: $C(13,1) \cdot C(4,3) \cdot C(12,2) \cdot C(4,1) \cdot C(4,1)$
- How many hands contain 2 pairs (e.g., 2 aces and 2 kings - the 5^{th} card cannot be ace or king):
- Number of ways to select 2 pairs is C(13,2).
- Number of ways to select suites of the 2 pairs $C(4,2) \cdot C(4,2)$.
- Number of ways to select fifth card: 52-8=44 (e.g., 5^{th} card different from 2 pairs).
- Total number: $C(13,2) \cdot C(4,2) \cdot C(4,2) \cdot 44 = C(13,2) \cdot C(4,2) \cdot C(4,2) \cdot C(11,1) \cdot 4$